

Network Security Analysis

Defensive Wireshark analysis of a controlled network traffic capture

1. Objective

Analyze captured network traffic, identify protocols and communication patterns, note security concerns, and provide defensive recommendations.

2. Capture Summary

Item	Result
Tool	Wireshark
Capture file	network_security_analysis.pcapng
Total packets	1,033
Dropped packets	0
Local IPv4	192.168.43.120
IPv4	693 packets (67.1%)
IPv6	333 packets (32.2%)
DNS filter	71 displayed packets
TLS filter	263 displayed packets
HTTP filter	4 displayed packets
TCP conversations	69

3. Methodology

Protocol Hierarchy, DNS, TLS, HTTP, IPv4 endpoints, and TCP Conversations were reviewed in Wireshark. The analysis is limited to the captured sample.

4. DNS Analysis

Observed domains included edge.microsoft.com, api.genuine.autodesk.com, and update.googleapis.com. These appear consistent with normal Microsoft, Autodesk, and Google-related services. No clearly malicious DNS pattern was identified.

5. TLS / HTTPS Analysis

TLSv1.2 traffic was observed, including encrypted Application Data over TCP port 443. This is consistent with secure web/application communication. Encrypted payload contents were not directly readable.

6. HTTP Analysis

Four HTTP packets were observed, including GET /connecttest.txt and HTTP/1.1 200 OK responses over port 80. This appears consistent with an internet connectivity check.

7. Security Concern

Plain HTTP is unencrypted and should not be used for sensitive information. HTTPS/TLS should be preferred. The observed connectivity-check traffic itself was not considered clearly malicious.

8. Endpoint Analysis

The local host 192.168.43.120 accounted for 693 IPv4 packets and approximately 325 kB. It transmitted 379 packets (about 209 kB) and received 314 packets (about 116 kB).

9. TCP Analysis

The TCP Conversations view contained 69 conversations. Most visible external connections used destination port 443, consistent with HTTPS/TLS. No clearly malicious high-volume TCP pattern was identified.

10. Findings

Normal DNS activity and predominantly encrypted TLS/HTTPS traffic were observed. A small amount of plain HTTP was the main security improvement opportunity. No clearly malicious pattern was identified in this limited capture.

11. Recommendations

Use HTTPS/TLS for sensitive communication; monitor unusual DNS queries; keep systems and applications updated; investigate unexpected endpoints with trusted threat-intelligence sources; and use Wireshark statistics/filters during future defensive investigations.

12. Conclusion

The sample was primarily consistent with normal network activity. The main observed security concern was limited unencrypted HTTP traffic. This conclusion applies only to the captured traffic and does not guarantee that the wider system or network is completely secure.